Semester Project Description
NE 511

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Abstract

This report introduces the Nonlinearly Implicit Low-Order CMFD (NILO-CMFD) method and the Multilevel Nonlinear Projective (MNP) method for solving multiphysics problems involving coupled neutron transport and Thermal Hydraulics. Both methods could be applied to any TH solver, but the MNP method, like X-CMFD, requires repeated inversion of the TH operator, which may potentially very computationally expensive. For my NE511 computational project I propose a NILO-MNP method which combines the advantages of these existing methods.

For the project, I will at minimum implement the MNP method, and if time allows I will implement the NILO-MNP method, and perform Fourier stability analysis on both methods.

1 Introduction

The NILO-CMFD method uses nonlinear diffusion acceleration (NDA) with coarse-mesh finite difference (CMFD) to acclerate coupled transport and TH. It also uses an approximate TH operator within inner iterations to reduce computational cost without introducing approximation in the final result.

The MNP method uses Multigroup and Grey Quasidiffusion to accelerate the same class of problems. In single-physics applications, Multilevel Quasidiffusion converges faster than NDA, but Fourier analysis has not been performed for the multiphysics case of MLQD.

This report summarizes the NILO-CMFD and MNP methods for coupled transport and TH. For my NE511 computational project I plan to implement the MNP method, first in the steady-state case, then with time-dependence and delayed neutrons.

The MNP method requires repeated evaluation of the full-order TH operator during inner iterations, which the NILO-CMFD method does not. If time allows, I would like to implement the MNP method using the same approximate TH operator in inner iterations.

I also plan to perform Fourier stability analysis to determined expected rate of convergence of the MNP method.

2 Problem Statement

The system of equations for 1D time-dependent multigroup neutron transport with delayed neutron precursors and thermal-hydraulic (TH) feedback is

$$\begin{aligned}
&\left[\frac{1}{v_g}\frac{\partial}{\partial t} + \mu\frac{\partial}{\partial z} + \Sigma_t(z)\right]\psi_g(z,\mu,t) \\
&\qquad = (1-\beta)\frac{\chi_g^p(z,t)}{2}\sum_{g'=1}^{N_g}\nu_g(z)\Sigma_{f,g'}(z)\phi_{g'}(z,t) + \frac{1}{2}\sum_{g'=1}^{N_g}\Sigma_{s,g'\to g}(z)\phi_{g'}(z,t) \\
&\qquad\qquad + \frac{1}{2}\sum_{k=1}^{6}\lambda_k\chi_{g,k}^d(z,t)C_k(z,t) + q_g(z,\mu,t), \\
&\qquad\qquad\qquad g = 1,\ldots,N_g,\ 0 \le z \le Z,\ -1 \le \mu \le 1,\ t > 0
\end{aligned} \tag{1a}$$

$$\psi_g(z,\mu,0) = \psi^0, \tag{1b}$$

$$\psi_g(0,\mu,t) = \psi_{in,g,0},\ \mu > 0 \qquad \psi_g(Z,\mu,t) = \psi_{in,g,Z},\ \mu < 0 \tag{1c}$$

$$\frac{\partial}{\partial t}C_k(z,t) = -\lambda_k C_k(z,t) + \sum_{g=1}^{N_g}\beta_{k,g}(z,t)\nu_g\Sigma_{f,g}(z)\phi_g(z,t), \qquad k = 1,\ldots,6 \tag{1d}$$

$$C_k(z,0) = C_k^0, \qquad k = 1,\ldots,6 \tag{1e}$$

$$\phi_g(z,t) = \int_{-1}^{1} d\mu\,\psi_g(z,\mu,t), \tag{1f}$$

$$\mathcal{H}T(z,t) = W\sum_{g=1}^{N_g}\Sigma_{f,g}(z)\phi_g(z,t), \tag{1g}$$

$$T(z,0) = T^0, \tag{1h}$$

where

z = spatial position

μ = direction cosine

t = time

v_g = particle velocity in group g

ψ_g = angular flux in group g

ϕ_g = scalar flux in group g

$\Sigma_{u,g}$ = cross section of reaction u

ν_g = fission multiplicity in group g

χ_g^p = prompt neutron energy emission spectrum

$\chi^d_{g,k}$ = delayed neutron energy emission spectrum from precursor group k

β = delayed neutron fraction

C_k = precursor population of group k

λ_k = average precursor decay constant of group k

T = temperature

W = energy released per fission

The TH operator $\mathcal{H}$ can be treated using any TH code, but for this application $\mathcal{H}$ will be the time-dependent advection-diffusion operator,

$$\mathcal{H}T(z) = \left[\frac{\partial}{\partial t} - \frac{\partial}{\partial z}k_b\frac{\partial}{\partial z} + \dot{m}c_p\frac{\partial}{\partial z}\right]T(z), \tag{2}$$

where

c_p = specific heat capacity

k_b = turbulent diffusion coefficient

$\dot{m}$ = mass flow rate

Similarly, cross-section temperature dependence may be treated using tabualted data or a general model accounting for material density, etc. For this application the temprature dependence of cross sections will be the model used in [1],

$$\Sigma_u(z) = \Sigma_{u,0} + \Sigma_{u,1,d}\left(\sqrt{T(z)+\kappa W h(z)} - \sqrt{T_{d,0}}\right) + \Sigma_{u,1,b}\left(T(z) - T_0\right) \tag{3}$$

where $h(z)$ is the fission reaction rate $\sum_{g=1}^{N_g}\Sigma_{f,g}\phi_g$ and $\kappa, \Sigma_{u,0}, \Sigma_{u,1,d}, \Sigma_{u,1,b}, T_{d,0}$, and T_0 are material constants, defined for each neutron energy group g.

The transport equation eq. (1a) and TH equation eq. (1g) can be written in operator form as

$$\frac{\partial\boldsymbol{\psi}}{\partial t} + \mathcal{L}[T]\boldsymbol{\psi} = \mathcal{S}[T]\boldsymbol{\psi} \tag{4}$$

$$\frac{\partial T}{\partial t} + \mathcal{H}T = \mathcal{Q}[T, \boldsymbol{\psi}] \tag{5}$$

where $\mathcal{L}$ represents streaming and collision, $\mathcal{S}$ represents source from scatter, fission, and precursor decay, and $\mathcal{Q}$ represents fission power hW.

Equations (1), (2) and (3) form a system of equations which are nonlinearly coupled and characterized by multiple time and length scales. The high dimensionality of eq. (1a) largely determines the complexity of the problem.

A standard approach to solving a coupled problem like this one would be Fixed-Point Iteration (FPI), in which an estimate $T^{(n)}$ is used to calculate $\psi^{(n+1)}$ using the transport equation. $\psi^{(n+1)}$ is then used to define the fission power distribution for a TH calculation which determines $T^{(n+1)}$. This procedure is repeated until convergence. FPI can converge very slowly in the case of tight coupling, and requires both a high-order transport calculation and a potentially costly TH calculation at each iteration.

The problem may also be solved using Newton's method or Jacobian-free Newton-Krylov methods, which still require solution of the high-order transport equation at each iteration.

The solution of the transport equation may be aided with a synthetic acceleration method, and the coupling may be aided using a low-order approximation of the TH operator. This is the basis of the NILO-CMFD method, which converges at a comparable rate to single-physics problems, and introduces no approximation to the final solution.

The MNP method [2] uses the Quasidiffusion method to project the transport equation to the GLOQD equation, which depends only on space and time. The coupled system of the MLOQD equations, precursor balance, and TH is then solved with Newton iteration.

3 NILO-CMFD Method

The NILO-CMFD method is described in [1] and [3]. It describes the method only for one-group, steady-state transport (without delayed neutrons), but the method can be easily extended for time-dependent calculations, since backward Euler discretization yields an equivalent steady-state problem (with augmented absorption cross section and source).

Each outer iteration of the CMFD-based methods consists of:

1. High-Order transport sweep using $\Sigma(T^{(n)})$
2. Calculation of corrected diffusion coefficient
3. Low-order diffusion solve for $\phi^{(n+1)}$
4. TH solution for $T^{(n+1)}$

The methods differ in how the low-order diffusion equation is solved:

- Fixed-point iteration solves the low-order equation using explicit cross sections (the same cross sections that were used for the high-order equation). FPI does not guarantee stability
- R-CMFD uses a "relaxed" heat source, where $h = \Sigma_f(\alpha\phi^{(n+1)} + (1-\alpha)\phi^{(n)})$. R-CMFD converges more slowly than FPI and may still be unstable for optically thick media.
- X-CMFD uses implicitly evaluated cross sections, requiring another level of FPI (or Newton's method) to solve the coupled system of the low-order diffusion equation and the TH equation. The method is stable, and converges at a rate comparable to single-physics transport, but the method requires multiple inversions of the TH operator $\mathcal{H}$ in each outer iteration.
- NILO-CMFD uses an approximation of temperature in the X-CMFD's coupled system of equations: cross section temperature dependence is linearized about the cross section values from the beginning of the iteration, and an approximation to the TH operator

$\mathcal{H}$ is used. The method converges at nearly the same rate as X-CMFD at a reduced computational cost, and introduces no approximation in the result.

3(a) NILO-CMFD

Algorithm 1 NILO-CMFD

Input: Initial iterates $T^{(0)}$ and $\phi^{(0)}$
  $\Sigma_u^{(0)} \leftarrow \Sigma_u(T^{(0)})$
  $n \leftarrow 0$
  while $\left\|T^{(n+1)} - T^{(n)}\right\|_\infty / \left\|T^{(n)}\right\|_\infty > \epsilon_n$ **do**
    Outer iterations;
    Calculate $\phi^{(n+1/2)}$ ▷ High-order transport sweep
    TH calculation for $T^{(n+1/2)}$
    Cross section evaluation for $\Sigma_u^{(n+1/2)}$
    Cross section evaluation for $\dot{\Sigma}_u^{(n+1/2)}$ ▷ Calculated numerically
    $s \leftarrow 0$
    while $\left\|T^{(s+1)} - T^{(s)}\right\|_\infty / \left\|T^{(s)}\right\|_\infty > \epsilon_s$ **do** ▷ Or use 'temperature update' convergence
      $s \leftarrow s + 1$
      Calculate $\phi^{(n+1,s+1)}$ using $\tilde{\Sigma}_u^{(n+1,s)}$ ▷ Low-order (Corrected Diffusion)
      Calculate approximate temperature update from $\tilde{\Sigma}^{(n+1,s)}\phi^{(n+1,s+1)} - \tilde{\Sigma}^{(n)}\phi^{(n)}$
      Update cross sections $\tilde{\Sigma}^{(n+1,s+1)}$ from Taylor expansion
    end while
    $\phi^{(n+1)} \leftarrow \phi^{(n+1,s+1)}$
    Update temperatures using full-order HT operator
  end while

Inside the inner iterations of the NILO-CMFD method, the implicit cross section temperature and flux dependence are estimated as a first-order Taylor expansion:

$$\begin{aligned}
\Sigma_u^{(n+1)} &= \Sigma_u(T^{(n+1)}, f^{(n+1)}) && (6)\\
&= \Sigma_u\left(T^{(n)} + (T^{(n+1)} - T^{(n)}), f^{(n)} + (f^{(n+1)} - f^{(n)})\right) && (7)\\
&\approx T^{(n)} + \frac{\partial \Sigma_u}{\partial T}^{(n)}(T^{(n+1)} - T^{(n)}) + \frac{\partial \Sigma_u}{\partial f}^{(n)}(f^{(n+1)} - f^{(n)}), && (8)
\end{aligned}$$

and $T^{(n+1)} - T^{(n)}$ is approximated using an approximate operator $H \approx \mathcal{H}$:

$$
\begin{aligned}
\Sigma_u^{(n+1)} &\approx T^{(n)} + \frac{\partial \Sigma_u}{\partial T}^{(n)} \mathcal{H}^{-1} W (\Sigma_f^{(n+1)} \phi^{(n+1)} - \Sigma_f^{(n)} \phi^{(n)}) \qquad (9)\\
&\qquad + \frac{\partial \Sigma_u}{\partial f}^{(n)} (\Sigma_f^{(n+1)} \phi^{(n+1)} - \Sigma_f^{(n)} \phi^{(n)})\\
&\approx T^{(n)} + \frac{\partial \Sigma_u}{\partial T}^{(n)} H^{-1} W (\Sigma_f^{(n+1)} \phi^{(n+1)} - \Sigma_f^{(n)} \phi^{(n)}) \qquad (10)\\
&\qquad + \frac{\partial \Sigma_u}{\partial f}^{(n)} (\Sigma_f^{(n+1)} \phi^{(n+1)} - \Sigma_f^{(n)} \phi^{(n)}).
\end{aligned}
$$

As the outer iterations converge, the term $(\Sigma_f^{(n+1)} \phi^{(n+1)} - \Sigma_f^{(n)} \phi^{(n)})$ goes to zero and thus there is no approximation in the final result. The approximate operator H should be chosen such that that it is accurate when T has weak spatial variation. [3] proposes H to be the advection operator without diffusion.

This method requires that

- Σ_f and ϕ must be stored from the previous iteration
- Two additional cross section evalutations are performed for numerical differentiation

The NILO-CMFD algorithm for a steady-state problem (or a single time step of a time-dependent problem) is presented in Algorithm 1.

4 MNP Method

A sketch of the Multilevel Nonlinear Projective method is presented in Algorithm 2. The full algorithm is presented in [2], which also gives the Multigroup Low-Order Quasidiffusion (MLOQD) and Grey LOQD (GLOQD) equations which are necessary for its implementation.

5 Preliminary and expected results

My implementation of the MNP method will use the Linear Discontinuous (LD) finite element discretization in 1D slab geometry, and finite volume discretization of the TH operator (for which I will use the advection-diffusion equation, as in [3]).

I have an LD transport solver which I can adapt to the high-order solver for this implementation; I expect to have that done this week. Afterwards, I will need to write the low-order solvers for MLOQD and GLOQD, and add a function for temperature-dependent cross section evaluations. The primary source of work for this early stage of implementation will be the finite volume advection-diffusion solver.

Once I have all solvers in place, I will just need to configure them for each of the methods, which should not be a challenge, as they all use the same kernel. For the Newton iterations in the MNP method I will use Jacobian-free Newton-Krylov iteration, which will introduce another level of iteration to Algorithm 2.

Algorithm 2 MNP

Input: Initial iterates $T^{(0)}$ and $\phi^{(0)}$

$n \leftarrow 0$

while $\left\|T^{(n+1)} - T^{(n)}\right\|_\infty / \left\|T^{(n)}\right\|_\infty > \epsilon_1$ **do**

  Outer Iterations;

  $n \leftarrow n + 1$

  Calculate multigroup cross sections

  Calculate effective grey cross sections

  Calculate $\psi^{(n+1)}$ using $\phi^{(n)}$; iterate through groups only once ▷ High-order transport

  Calculate $E_g^{(n+1)}$, $B_g^{(n+1)}$

  $s \leftarrow 0$

  while $\left\|T^{(n,s+1)} - T^{(n,s)}\right\|_\infty / \left\|T^{(n,s)}\right\|_\infty > \epsilon_2$ **do**

    Multigroup QD iterations;

    $s \leftarrow s + 1$

    Update cross sections, fission power source

    Solve MLOQD equations for $\phi^{(n+1,s+1)}$ and $J^{(n+1,s+1)}$

    Calcualte effective grey closure terms and cross sections

    $l \leftarrow 0$

    while $\left\|T^{(n,s+1,l+1)} - T^{(n,s+1,l)}\right\|_\infty / \left\|T^{(n,s+1,l)}\right\|_\infty > \epsilon_3$ **do**

      Newton iterations;

      $l \leftarrow l + 1$

      Solve the system GLOQD, $\Sigma(T)$, and linearized full-order TH.

    end while

  end while

end while

A time-dependent transient benchmark is introduced in [2], which I can compare my results to. If I implement both the MNP and NILO-MNP methods, I will record the number of each kind of iteration required (i.e. to count number of $n \times n$ linear solves required for each method) and residual histories for each method to show convergence rate.

If I am able to perform Fourier stability analysis, the intended results are:

- Region of stability for the NILO-MNP method
- Spectral radii for FPI-MNP, MNP, and NILO-MNP methods

References

[1] N. J. Adamowicz, A. Manera, and E. W. Larsen, "The NILO-CMFD method for iteratively solving coupled neutron transport–thermal hydraulics problems," *Nuclear Science and Engineering*, vol. 197, no. 2, pp. 262–278, 2023.

[2] A. Tamang and D. Y. Anistratov, "A multilevel projective method for solving the space-time multigroup neutron kinetics equations coupled with the heat transfer equation," *Nuclear Science and Engineering*, vol. 177, no. 1, pp. 1–18, 2014.

[3] "The NILO-CMFD method for iteratively solving multiphysics neutron transport - thermal hydraulic problems," in *Proceedings of the American Nuclear Society Conference on Mathematics and Computation 2021*, pp. 2092–2101, ANS, 2021.